SIH 2026 · PS 26147 · SPONSOR NTRO · THEME SPACE TECHNOLOGY · CATEGORY SOFTWARE

Project SANJAYA

Slide-wise content for the idea-round PPT — built on the 6-WH framework: **What · Why · When · Where · Who · How**

How to use this document. Each numbered block below is one slide of the deck: the slide number, which WH it answers, the slide headline (use it verbatim - judges skim headlines), and the body content in speakable bullets. Every number is from the team's own documents and measured evidence (PRD v2.4, engine build state of 27 Sep 2026, competitor snapshot of 26 Sep 2026) - nothing is invented, so nothing collapses under a judge's question. The deck is 16 slides: 1 title + 12 content + 2 backup + 1 close, comfortably fitted to a 5-7 minute pitch.

WH	Slides	One-line job of those slides
WHAT	2-5	The problem, our solution, what we built already, and the differentiator (the proof).
WHY	6-7	Why it matters to NTRO, why now, and the quantified field gap (43 repos analysed).
WHEN	8	A dated, confidence-backed execution plan with gates.
WHERE	9	Deployment inside NTRO's air-gapped reality + who benefits.
WHO	10-11	End-users and the six-member team.
HOW	12-15	Architecture, tech stack, intelligence, data strategy, feasibility already measured.
CLOSE	16	The three things judges must remember.

SLIDE 1 COVER

Project SANJAYA — the tool that proves its answers

Team name · PS 26147 · NTRO · Space Technology · Software

Headline on the slide: “Automated analysis of .IQ and .wav recordings — every answer ships with a verifiable proof.”

- **Subtitle:** An offline, air-gap-ready system that turns any .IQ / .wav recording into a complete, **proven** transmitter recipe: sampling frequency, modulation, FEC, interleaving, scrambling, frame structure, header and payload.
- **Tagline strip at the bottom (3 chips):** “Be right, or say you’re unsure” · “A certificate anyone can re-check in seconds” · “Built already: engine + UI running end to end”.
- **Footer facts (small print, they pre-empt judge doubts):** PS 26147 · 100 % of PS clauses R-01–R-17 in scope · 5 crown innovations · key compute and proof claims already measured · working code, not slideware.

SLIDE 2 WHAT

The problem: an analyst gets a recording — and a guess

PS 26147 asks for the full automated chain; today the answer arrives without a way to trust it

- **The PS, in its own words:** a GUI-based model takes a .IQ or .wav file and must (i) identify signal parameters — sampling frequency, modulation, FEC, interleaving; (ii) demodulate FSK / QAM / PSK; (iii) de-interleave (block, convolutional, diagonal, pseudo-random); (iv) FEC-decode (Viterbi convolutional, RS, concatenated, LDPC); (v) bit-stream correlation to find header and payload.
- **The PS names the pain itself:** data is “often insufficient for fine-grain analysis”, it comes “from different sensors and different locations”, and this “reduces the accuracy and confidence of interpretation”. We answer each complaint with a named capability — not a claim, a mechanism.
- **Why it is genuinely hard:** blind — no metadata, unknown Fs; deeply ambiguous — a wrong code can “correct” its own errors and look perfect; and real recordings carry multiple emitters, bursts and HF fading. Every blind tool answers this question; almost none can be **checked**.
- **Our one-line problem statement:** “Give the analyst not an answer, but a **proven** answer — and an honest ‘not yet decidable’ when the data does not support one.”

Slide design cue: show a waterfall image with a red “?” → the question every analyst asks: “What is this, and can I trust what the tool says?” Keep text minimal; the 4 bullets above are speaker notes.

SLIDE 3 WHAT

Our solution: SANJAYA — analysis by synthesis, with a layered proof

Decode → re-synthesise → subtract → certify layer by layer → sign a certificate

- **The loop:** OBSERVE → MEASURE → HYPOTHESISE → DEMODULATE → UNWRAP → STRUCTURE → PROVE → LEARN. A proof FAIL loops back into the hypothesis tree; a PASS issues the certificate.
- **1. Decode.** Hypothesis-tree search over emitters → modulation → interleaver → code → scrambler → framing, with ambiguity hypotheses (phase rotations, conjugation) carried explicitly.
- **2. Re-synthesise & subtract.** Rebuild the waveform from the decoded bits through the recovered chain and subtract it from the recording → the residual.
- **3. Prove each layer with a test that has power at that layer:** PHY — residual at the measured noise floor (equivalence tests, not idealised thresholds); CODING — parity-membership test K1-K3; STRUCTURE — sync, CRC, counter checks. Chance of faking CRC: 2^{-16} per frame; 20/20 CRC-16 passes by luck = 2^{-320}.
- **4. Sign it.** An Ed25519-signed **Decode Certificate**: input SHA-256, recipe, per-layer verdicts, every statistic, near-misses rejected, K = number of proof attempts and its false-PASS bound.
- **5. Verify independently. sanjaya verify file cert.json** re-runs the proof from the file alone — no search, no trust, seconds. Anyone with the file and the public key can re-check any analyst's certificate.

2M+
near-miss proof attempts planned to bound false certificates

≤10^{-4}
wrong-certificate bound per certificate (both fast & deep paths)

~1.7 s
measured independent re-verification of a signed certificate

SLIDE 4 | WHAT

This is not slideware: the engine and UI already run end to end

Status as of 27 Sep 2026 — measured on the team's own machine, seed 147

- **Working engine (Python, G1 vertical slice + G2 blind search):** drop a SigMF / WAV / raw-IQ file → **blind** decode in 2-13 s: symbol rate, roll-off, CFO, clock ppm, frame + sync, block interleaver geometry, rate-1/2 convolutional code (K 3-9), frame-synchronous scrambler (library + exhaustive ≤ degree 8), phase ambiguity, counters, CRC (polynomial by GF(2) GCD).
- **Proof already runs:** layered proof PASS (PHY T1-T4 vs measured noise reference, coding K1/K2, structure S1-S4) on a Gen v0 file: 40/40 frames, 0 info-bit errors at 12 and 8 dB; CFO error < 0.01 Hz; Ed25519-signed certificate; independent re-verification in ~1.7 s.
- **Working UI (design frozen):** React 19 + three.js 5-step flow — Recording → Signal → Decoding → Header & Payload → Proof (+ Certificate); real in-browser WAV/SigMF parsing, raw-IQ dtype sniffing, triage, bit autocorrelation / entropy / CRC search. Engine↔UI verified end to end in Chrome: drop → blind decode → proof residual at noise floor → signed certificate.
- **Real HF recordings, not just synthetic:** a one-click sample library in the UI (6 synthetic + 6 real off-air recordings from the Signal Identification Wiki — STANAG 4285, MIL-188-110A, HFDL, PSK31, SITOR-B, 2G ALE); band-edge and symbol/tone-rate measurement rewritten this week to the standard −20 dB bandwidth method — PSK31 rate now exact (31.25 Bd), SITOR-B rate exact with a clean 2-tone pair, ALE's 8-tone set matches the MIL-STD-188-141 plan to within a few Hz.
- **Honest on the recordings it can't yet decode:** when the blind search comes back undecided, the UI still shows the real spectrogram and every measured parameter, plus the reason — never a guess dressed up as an answer.
- **The four USPs are no longer aspirational — all four run today, on real recordings:** a physics-only classifier identifies rate/order/standard on all 6 real off-air files with no training data (STANAG 2399.92 Bd at −32 ppm, PSK31 31.25 Bd exact); the Data-Sufficiency Engine predicts "record 0.90 s (P50) / 1.06 s (P90)" for a too-short capture, and a capture cut to that exact length then decodes; Evidence Stitching pools three sensor captures (different ADC rates, clocks 0/−25/+40 ppm, one inverted) each too short alone into a decode with every CRC valid, catches a decoy from a different emitter, and reports each sensor's clock/tuning/gain; the Recipe Compiler turns a certified recipe into a dependency-free receiver.py, gated so it is only released once it reproduces SANJAYA's bits exactly — verified bit-exact and portable to a different sensor.
- **64 pytest tests** (layering rules, wrong-recipe, tampered-certificate, wrong-file, real-file measurement, sufficiency-prediction, stitching, compiler bit-exactness regression cases) — plus a nightly kill-matrix plan so **every fragile dependency has a proven fallback**.

Why this slide wins rooms: most teams pitch what they will build. We show a blind decode + signed proof + independent verify working **today**, with honest limits stated (schema 0.1: QPSK-only engine, proof thresholds pre-calibration) - space-agency honesty as a feature.

SLIDE 5 WHAT

Five differentiators no competitor even claims — and why the gap lasts

43 public competitor repositories analysed (README claims, snapshot 26 Sep 2026)

#	Where the field stops	SANJAYA — working today, not planned
1	One file at a time	Evidence Stitching — 3 sensor captures, each too short alone, pooled to a full decode; every CRC valid
2	One sensor at a time	Sensor Twin — clock −25.6/+37.4 ppm measured against the other sensors, tuning, gain, spectral inversion
3	No answer to “is this enough data?”	Data-Sufficiency Engine — predicts “record 0.90 s (P50) / 1.06 s (P90)”; a capture that long decodes
4	Output ends in a report	Recipe Compiler — receiver.py, gated bit-exact (40/40 frames), portable to another sensor
5	Bit-level verification at best	Layered Proof + signed Decode Certificate; verify-in-UI: drop a certificate + file, get REPRODUCED

- **33 of 43 competitor READMEs** test truth only against their own generator; SANJAYA validates on real public recordings with truth from **independent** decoders (SatDump, dumphfdl, fldigi).
- **Why the gap lasts:** the five capabilities depend on each other — stitching needs the Twin and the sufficiency maths; the compiler takes certified recipes only; the proof is the arbiter of both. Copying one idea does not reproduce the system.

SLIDE 6 WHY

Why NTRO should care: four PS complaints, four delivered capabilities

Every PS paraphrase on the left is answered by a mechanism, with a pre-registered KPI

PS complaint (paraphrased)	Our answer	Pre-registered target
Data “often insufficient for fine-grain analysis”	C2 Data-Sufficiency + D1 soft joint ID (needs ≤ 3N interleaver rows vs $2N/(1-p)^N$ hard-rank)	Decision rate ≥ 90 % in nominal envelope; correct-certified ≥ 90 %
Data “from different sensors and different locations”	C3 Stitching + C4 Sensor Twin (associate → normalise → align → pool log-likelihoods)	Alignment loss ≤ 10 %; false merge ≤ 1 %; validated on ≥ 3 KiwiSDR sites, GPS-timestamped
“Reduces accuracy and confidence of interpretation”	C1 layered proof + uncertainty on every value + conformal sets on classes	Wrong-certified ≤ 10□⁴ per certificate; P95 interval coverage 0.95 ± 0.03
Output “utilised in the designated sensors”	C5 Recipe Compiler — certified recipe → GNU Radio .grc + recipe JSON + SigMF	100 % bit-exact on the source recording, proven at the release gate

- **Deployable where NTRO actually works:** zero outbound network packets (tested by packet capture), 127.0.0.1-only API, offline per-user installer incl. offline WebView2, deterministic certificates, SBOM + pinned dependencies, fuzzed parsers — engineered for the air gap from day one.
- **Trust you can audit:** the certificate binds input SHA-256 + engine version + every statistic; a second machine can re-verify any verdict in seconds. **Analysts stop arguing about whose tool is right.**

SLIDE 7 WHY

Why now — the quantified field gap (43 repositories, ranked)

Snapshot 26 Sep 2026; strongest three competitors shown; README claims, verified not assumed

Capability	Top repo A (9.0)	Top repo B (8.8)	Top repo C (8.6)	SANJAYA
PS (i)-(v) full chain	Yes	Yes	Yes	**Yes**
GUI	Yes	Not evident	Yes	**Yes (built)**
Blind interleaver size	Hard rank	Hard rank	Catalogue	**Soft, joint (<3N rows)**
Proof beyond bit level	No	No	CRC gate only	**Layered proof**
Third party can re-check result	No	No	No	**Yes (verify)**
“Is this enough data?”	No	No	No	**Yes (C2)**
Pools captures & sensors	No	No	No	**Yes (C3/C4)**
Receiver for sensors	No	No	No	**Yes (C5)**
Real signal, full chain, independent truth	No	No	No	**Yes (Meteor vs SatDump)**
Blind scrambler ID incl. post-FEC	No	Fixed only	No	**Yes**

- **Why “now”:** the hard parts stopped being research and became engineering — our feasibility script **measured** the critical numbers (next slides). The gap above is structural, not incremental: five interlocking capabilities that cannot be copied one at a time.

SLIDE 8 WHEN

When: a 10-week, gate-driven plan — committed at 73 % schedule confidence

W1 = 28 Sep 2026; NASA-style threshold vs baseline; weekly earned value with mechanical descoping

Gate	Date	Exit criterion (what exists when we leave the gate)
G0	4 Oct	Engineering baseline: interfaces frozen, CI + 25-test check green, KPI definitions drafted
G1	11 Oct	Vertical slice: Gen v0 file blind-decoded, layered proof passes, API + visual-regression baseline
G2	25 Oct	Blind conv code + interleavers on synthetic; proof calibrated; first Meteor-M capture
G3	8 Nov	RS + post-FEC scrambler → **Meteor-M LRPT fully blind, frames equal SatDump**; FSK/MSK; triage
G4	22 Nov	LDPC, HF curves, bursts (HFDL), C2/C3/C5, offline installers; **all requirements pass**
G5	29 Nov	Independent KPI report on sealed sets; 2M-attempt false-PASS campaign done; zero-network test; freeze

- **Honest capacity maths:** 6 members × 25 h/week × 10 weeks = 37.5 pw capacity. Baseline plan = 34.35 pw (31 % confidence) is the **goal**; **threshold configuration = 31.65 pw committed at 73 % joint confidence** (200,000-run Monte Carlo, triangular −20 %/+50 % effort model) — NASA JCL-style commitment, not optimism.
- **Mechanical, pre-agreed descoping ladder** (8 rungs, 2.70 pw) protects the threshold: PS coverage and the layered proof are never cut. P2 stretch items start only if SPI ≥ 1.0 and ≥ 2.0 pw reserve remains.
- **Why judges should trust the timeline:** gates have dated, inspectable exit criteria; loading is capped at 1.25 pw per role per phase; the critical path (Gen → Viterbi/GF(2) → coding ID → proof → Meteor blind → C3/C5 → independent verification) is identified with zero float on coding and ML.

SLIDE 9 WHERE

Where it runs: NTRO’s air-gapped analysis workflow — end to end

One laptop, no GPU, no network — from intake to a receiver NTRO's sensors can run

- **In the lab:** any analyst workstation (8-core, 16 GB RAM, no GPU needed) — Windows 10/11 and Ubuntu 22.04/24.04; files to 8 GB memory-mapped; engine RAM ≤ 4 GB; first results in seconds, deep search anytime-capped.
- **In the field:** captures from **different sensors and locations** flow into C3 stitching — public KiwiSDR network receivers (multi-country, GPS-timestamped) for validation; SigMF as the interchange format.
- **At the sensors:** C5 hands the designated receiver a certified recipe (JSON) plus a self-contained GNU Radio flowgraph and a recipe-matched detector (+15 dB coherent gain for a 32-symbol sync word) — the PS’s “utilised in designated sensors” made literal.
- **Who benefits:** the **signal analyst** (proven answers, honest limits), the **collection manager** (tasking note: what to record, where, how long — P90 duration), the **sensor engineer** (ready-to-run receiver), and the **independent checker** (verify any certificate in seconds).

Real signals, real places (validation targets): Meteor-M N2-3/N2-4 LRPT at 137.9 MHz (full chain vs SatDump), HFDL bursts (vs dumphfdl), STANAG 4285 (PHY-level truth), NAVTEX (vs fldigi) — all public broadcasts; payloads never decrypted; KiwiSDR operators credited and respected.

SLIDE 10 WHO

Who uses it: four users, four jobs — designed around the analyst

Primary user: the expert signal analyst on an air-gapped workstation

User	Their job to be done (verbatim from our user research)
Signal analyst (primary)	“Tell me everything about this recording and show me why you believe it. Tell me when you can’t know.”
Collection manager	“Is this enough data? If not, what should I record, where and for how long?” — C2 answers with a P90 tasking note.
Sensor engineer	“Give my sensor a receiver for this signal.” — C5 compiles it, bit-exact.
Independent checker	“Let me check this certificate without trusting your search.” — one CLI command, seconds.

- **Design consequence:** every value carries its uncertainty (u and a 95 % interval); every class answer carries a calibrated conformal set; “undecided” is a **first-class output** with a data-needed note — never a confident wrong answer.

SLIDE 11 WHO

Who we are: six specialists, five roles, one review-driven culture

Replace bracketed names — keep the role structure; it matches the WBS 1:1

Role	Owns (work packages)	Proof-of-work already visible in the repo
DSP	WP02-03: measurement, receivers, bursts	Feedforward timing estimator measured at 0.011 s per 10^6 samples
Coding	WP05-06: C++ kernels, blind coding ID	GF(2) rank 2048×1024 in 0.069 s; blind conv-ID K 3-9 works
Protocol	WP07-08, WP13, WP17-20: Gen, Bit-Lens, orchestrator	Generator with per-layer taps; 42-test CI; blind search 2-13 s
ML	WP04, WP09-10: experts, proof, sufficiency	Layered proof PASS measured (K1: 0.0003 vs 0.494 random ≈ 88σ)
Product	WP12, WP15-16, WP21-22: UI, Tauri, installers	React + three.js UI built and engine-wired; certificate PDF/JSON
Data	WP01, WP11, WP14, WP18-19: ingest, real data, acceptance	SigMF/WAV/RAW ingest + dtype sniffing working in browser

- **Review-driven:** our own first PRD was put through a formal 44-finding design review (5 critical) and rebuilt to v2.4 - we grade ourselves harder than the judges will.
- **Independent verification inside the team:** sealed test sets via commit-reveal (SHA-256 published at G1, revealed at G5), run by members who did **not** build the module.

SLIDE 12 HOW

How it works: 11-layer engine under a frozen UI, one contract

Layered architecture, enforceable in CI; Pydantic → OpenAPI → TypeScript — one schema source

- **Stack:** Python 3.10 + NumPy/SciPy orchestration; **C++17 kernels via pybind11** for Viterbi, GF(2), RS, LDPC, tracking loops; FastAPI + WebSockets engine bound to 127.0.0.1; ONNX Runtime for every model; Tauri 2 desktop shell with offline WebView2; GNU Radio only as the C5 compiler target and test oracle.
- **Orchestrator:** best-first hypothesis-tree search (A*-style) with cheap calibrated pruning (wrong-branch pruning < 10□³ measured target); memoised nodes; fast path ≤ 8 proof attempts in < 10 s per 10^6 samples; deep path ≤ 64 attempts, anytime, capped at 10 min.
- **Compute budget, measured:** ingest+LOD 0.62 s for a 100 MB file (3× margin on the <2 s KPI); GF(2) rank 2048×1024 = 0.069 s; pure-Python tracking loop 0.25 s per 10^6 samples; NumPy Viterbi fallback 0.15 Mbit/s with 0 post-FEC errors at 6 dB — **every budget line has a measured or allocated value and a 150 % trip-wire**.
- **Safety & security:** per-job worker processes with OS memory caps and watchdogs; Host/Origin checks + per-launch token (anti DNS-rebinding); fuzzed parsers (target ≥ 24 CPU-h, zero crashes); Ed25519 key in the OS key store; zero runtime network calls, proven by packet capture.

SLIDE 13 HOW

How the intelligence works: learned experts propose, physics and algebra verify

A model error costs time — never correctness

- **Three experts:** E1 feature GBDT (cumulants, power lines, cyclic features); E2 compact CNN (< 3 M parameters, ONNX, rate-normalised inputs so the net always sees a fixed samples-per-symbol); E3 decode-fit likelihood — **physics immune to the sim-to-real gap**.
- **Fusion with trust:** per-expert temperature scaling, logistic stacking, split-conformal sets calibrated on real data (Mondrian + SNR-binned coverage, n ≥ 8,000 per tier), NOVEL detection by empty set.
- **Coding layer is deliberately algebra, not ML:** soft rank profiles, parity-membership tests, GF(2) CRC solving, Berlekamp-Massey scrambler recovery incl. post-FEC cosets (the Meteor-M case: CCSDS randomiser after RS, dual-basis RS(255,223), interleave depth 4).
- **10 modulation classes in P1:** BPSK, QPSK, OQPSK, 8PSK, 16QAM, 64QAM, 2/4/8-FSK, MSK; FEC: conv K ≤ 9 (punctured), RS both bases × interleave depth, LDPC catalogue (DVB-S2, CCSDS, 802.11, 5G NR), concatenated — 100 % of PS clause (iv).

SLIDE 14 HOW

How we beat the data problem: 4 tiers, 8 levers — and proof we need less data

No hardware needed: exact-truth synthetic + real public recordings with independent truth

- **Tiers:** T1 SANJAYA-Gen (exact truth, per-layer taps so every WP develops unblocked) → T2 Gen with measured-style front-end/channel models → T3 real public recordings (KiwiSDR, SatNOGS/SatDump samples) with truth from **independent decoders** → sealed commit-reveal test sets (SHA-256 published at G1).
- **Why it matters, with numbers:** hard-decision rank needs $2N/(1-p)^N$ error-free rows — at N = 1,024 that is **44,411 rows (≈ 6.9× our 3N target of 3,072)**; our soft joint ID targets ≤ 3N rows. This is where the PS's "insufficient data" complaint is actually answered.
- **Real signals with independent truth:** Meteor-M LRPT 137.9 MHz (full chain vs SatDump), HFDL bursts (vs dumphfdl), STANAG 4285 (PHY-only truth), NAVTEX (vs fldigi) — 33 of 43 competitor READMEs never leave their own generator; our Tier-3 KPIs are separate and reported per tier.
- **Test integrity:** commit-reveal sealed sets; held-out parameter regions, held-out sensors, a held-out signal family; independence (non-builders run sealed sets); every KPI reported with Wilson/coverage intervals.

SLIDE 15 HOW

Feasibility is measured, not promised — key claims already proven

From _build/feasibility.py, run 27 Sep 2026 on the team laptop; seed 147

Claim	Measured result	Verdict
100 MB file interactive < 2 s (R-11)	All LOD tiles, 25 M samples: **0.62 s**	Holds, 3× margin
Parity test separates coded from random	f = 0.0003 coded vs 0.494 random (≈ 88σ)	Decisive at both SNRs
PHY residual vs measured noise reference	0.002 dB at 12 dB; 0.043 dB at 6 dB	Equivalence margins hold
Blind search on real files	2-13 s per file, full recipe, 40/40 frames	Works end to end
Independent re-verification	~1.7 s from file + certificate alone	Checking is cheap
Schedule confidence (200k Monte Carlo)	Threshold 31.65 pw: **73 %**; baseline: 31 %	Commit at threshold
False-PASS bound	2M near-miss attempts → ≤ 1.5×10□⁶/attempt × 64 = 9.6×10□⁵	≤ 10□⁴ per certificate

- **What is still unmeasured is stated too** (C++ kernel speed, recognition/proof stage times, real-signal behaviour) - each has an allocation, a gate date and a 150 % trip-wire. **We say what we do not know.**

SLIDE 16 CLOSE

If you remember three things

- **1. We prove our answers.** The only entry whose result ships as an Ed25519-signed, layered Decode Certificate that anyone can re-verify in seconds (≤ 10□⁴ wrong-certified bound; ~1.7 s measured re-verification).
- **2. We already run.** Blind decode + layered proof + independent verify working end to end today (engine + frozen-design UI), with key compute claims measured (0.62 s / 100 MB; blind search 2-13 s).
- **3. We answer the PS's complaints with mechanisms, not adjectives** — sufficiency (C2), multi-sensor stitching (C3/C4), confidence (C1), and a ready-to-run receiver for designated sensors (C5).

Ask: selection to the next round. **Then:** G2 on 25 Oct — first Meteor-M capture decoded blind against SatDump's frames. Watch that space.

Backup slides (prepared, shown only if asked)	
B1	Full KPI table with acceptance decision rules (S/P/C/E classes) — PRD §3.4-3.5
B2	Head-to-head vs all 43 repos + scoring method — snapshot 26 Sep 2026
B3	Risk register (top 11 scored risks with triggers) — PRD §12.3
B4	Live demo script with per-beat fallbacks — PRD §12.4
B5	Requirement traceability R-01–R-17 → WP → test → KPI → demo beat — PRD §5.3

Sources: PRD.md v2.4 (KPIs §3, USPs §4, plan §11, evidence §16) · HANDOFF.md 27 Sep 2026 (build state) · _build/data.json (43-repo snapshot) · _build/feasibility.py (measurements). All KPIs are pre-registered targets; competitor statements are README claims.